

Nikhil Bansal

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EDUCATION

University of North Carolina at Charlotte

Master of Science in Computer Science | GPA: 4.0/4.0

Charlotte, North Carolina, USA

August 2022 - December 2023

Acropolis Institute of Technology and Research

Bachelor of Technology in Computer Science Engineering | GPA: 3.23/4.0

Indore, Madhya Pradesh, India

June 2017 - 2021

SKILLS

Languages and Cloud Skills: C/C++, JAVA, Python, Golang, JavaScript, Kubernetes, Docker, AWS (S3, DynamoDB, Cognito), AzureAD, Firebase

Web & Backend: Spring Boot, REST APIs, gRPC APIs, MongoDB, MySQL, SQL, Node.js, Express.js, HTML, CSS, Bootstrap

Tools & Platforms: JupyterHub, JupyterLab, Git, GitHub, Bitbucket, JIRA, Bamboo, Splunk, MS Office Suite

Other: Data Structures & Algorithms, Object-Oriented Programming, Debugging, MacOS/Windows/Ubuntu

EXPERIENCE

Intel Corporation | Cloud Software Development Engineer

February/12/2024 - Present | Santa Clara, California, USA

- Led PoC to production rollout of an AI-ready JupyterHub platform, providing on-demand access to 30+ Intel GPU and Habana Gaudi nodes for running 23+ GenAI and AI models, including LLMs, text-to-image, image-to-image, DeepSeek, and PyTorch
- Enabled users to land in custom container-based environments with persistent storage and pre-installed model-specific dependencies.
- Developed features include: MetalLB load balancer, NFS dynamic volume provisioning, Azure AD AuthN/AuthZ, Microservice for user registration, model-based callback routing, rate limiting, and Coupon-based access system for premium model usage.
- Designed and integrated custom Dockerfiles stored in AWS ECR, enabling dynamic image selection in pods for user model selection.
- Authored comprehensive design documentation, architectural diagrams, and recovery plans; led stakeholder presentations and obtained approval through Architecture Board Review (ABR) and SAFE processes.
- Added Prometheus for monitoring, Nessus for security scanning, and enforced network policies using Calico and Istio
- Developed custom JupyterLab extensions to enhance user experience (e.g., live session timers, announcement popups, error handling UI), and implemented intelligent node and GPU card scheduling for optimal resource utilization.

Intel Corporation | Graduate Technical Intern

May/15/2023 to December/15/2023 | Santa Clara, California, USA

- Establish JupyterHub architecture solution using Slurm, dynamic user creation with webhooks, and Azure AD authentication.
- Design and implement multiple backend APIs with database architecture for oneAPI Intel Developer Cloud web services.
- Developed a cron-job scheduler to provision various instances in a Kubernetes cluster and user requests every 30 seconds.
- Fashioned complex SQL queries to retrieve and manipulate data with future enhancement use cases. (DDL & DML queries)

University of North Carolina at Charlotte | Research Assistant

October/10/2022 to April/15/2023 | Charlotte, North Carolina, USA

- Worked on a Mobile app that leverages AI technology to collect valuable information from users using mobile sensors, which is then utilized to offer personalized recommendations for enhancing their workout routines.
- Developed numerous E2E solutions, from establishing system needs and partner dependencies to workload balancing, software implementation, testing, and configuring metrics, alarms, monitors, and dashboards using JAVA, JavaScript, and AWS.
- Devised a custom DynamoDB database and API library for integration with the data management application.
- Established a secure authentication system using AWS, ensuring data privacy and user confidentiality.

Tata Consultancy Services Ltd. | Assistant System Engineer

June/03/2021 to July/22/2022 | Indore, Madhya Pradesh, India

- Transform the appointment scheduler application from old generation Architecture to new generation MVC Architecture.
- Formulated and innovated Cloud Formation Template for database schema generation in AWS for production release.
- Demonstrated exceptional proficiency by designing and establishing multiple REST APIs from scratch to seamlessly connect microservices with the database, optimizing the application's performance and functionality.

Deqode Techracers Pvt. Ltd. | Solution Engineer Intern

January/04/2021 to May/04/2021 | Indore, Madhya Pradesh, India

- Executed geolocation functionality using the Google Maps API, enhancing user experience with accurate location-based services.
- Exhibited expertise in building multiple UI layouts and optimizing UI event handling by skillfully utilizing the main thread and subsidiary activities through worker threads, resulting in a seamless and responsive user interface.
- Utilized data parsing and data transfer between multiple fragments using a single activity for performance optimization, reducing the average request processing time by 74% from 4.8 to 1.1 sec.

PROJECTS & CERTIFICATIONS

- **Heal App [JAVA]** - Platform to help people reach online meditation, yoga, and music using mobile sensors.
"Importance of Mediation" (ISSN: 2582-3930) | International Journal of Scientific Research in Engineering and Management (IJSREM)
- **Travelers [Node.js, Express.js, MongoDB]** - An authentication-based e-commerce website with integrated payment gateways.
- AWS Academy Cloud Foundations, Scrum Fundamentals Certified (SFC), The Web Developer Bootcamp, Microsoft Technology Associate: Database Administration Fundamentals (MTA), Programming for Everybody (Getting Started with Python).